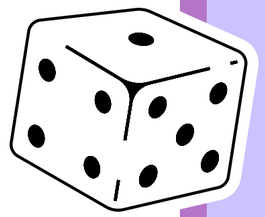
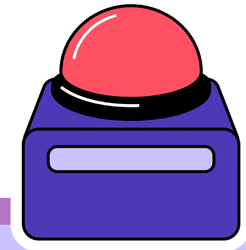
BCC1244 - GAME PROGRAMMING





3D GAME DEVELOPMENT

START





OVERVIEW

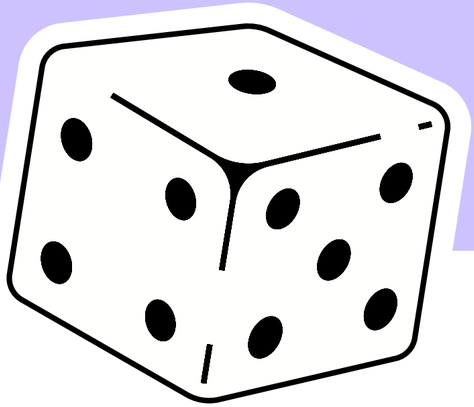
- *3D Game Development is the process of creating games that use three-dimensional graphics, environments, and objects.*
- *Involves modeling, texturing, animation, physics, and rendering in a 3D space.*



Applications of 3D Games

- *Entertainment (e.g., AAA games, indie games)*
- *Simulation (e.g., flight simulators, medical training)*
- *Virtual Reality (VR) and Augmented Reality (AR)*





CORE COMPONENTS OF 3D GAME DEVELOPMENT

CORE COMPONENTS

Game Engines

Software frameworks for building games.

Popular engines:

- *Unity: User-friendly, cross-platform, great for beginners.*
- *Unreal Engine: High-end graphics, used in AAA games.*
- *Godot: Open-source, lightweight, and flexible.*

Features: Rendering, physics, scripting, asset management, and more.





CORE COMPONENTS

3D Modeling and Assets

3D Models: Objects, characters, and environments created using tools like:

- *Blender, Maya, 3ds Max, ZBrush.*



Textures: 2D images applied to 3D models to add detail (e.g., colors, patterns).

Animations: Movement of characters and objects (e.g., rigging, keyframes).

CORE COMPONENTS

Physics and Collision Detection

Simulate real-world physics (e.g., gravity, collisions).

Tools: Built-in physics engines (e.g., Unity's PhysX, Unreal's Chaos Physics).



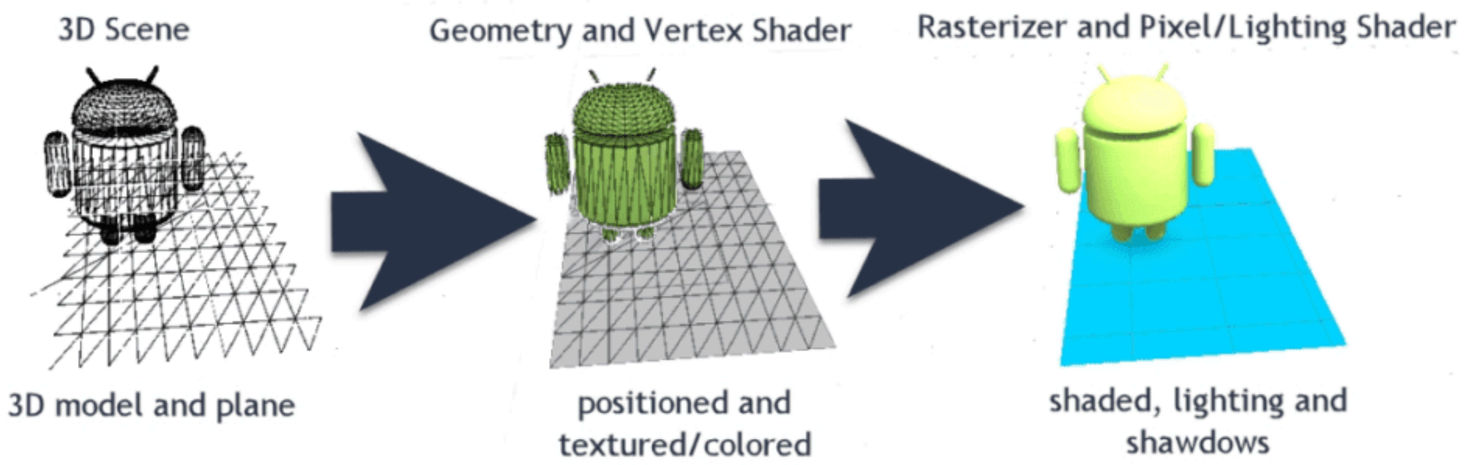
CORE COMPONENTS

Lighting and Rendering

Lighting: Simulates light sources (e.g., directional, point, spotlights).

Rendering: A process that uses computer software to create a 2D image from a 3D model

Techniques: Real-time rendering, ray tracing, global illumination.



CORE COMPONENTS

Audio

Sound effects, background music, and voiceovers.

Tools: FMOD, Wwise, or built-in audio systems in game engines.



Unity Learn | Foundations of Real-Time Audio | URP

4.5 (53)

UNITY TECHNOLOGIES

Free

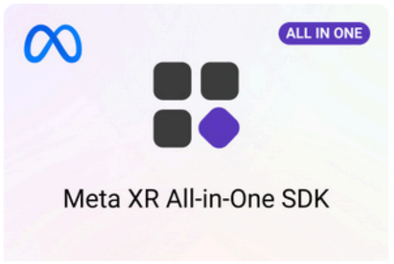


Bro Audio

5 (14)

AMI PRODUCTION

Free



Meta XR All-in-One SDK

2.5 (69)

META

Free



Audio Tools for Unity Playground

4.3 (6)

MELODRIVE

Free



FMOD for Unity (2.02)

3.8 (78)

FMOD

Free

VERIFIED SOLUTION

CORE COMPONENTS

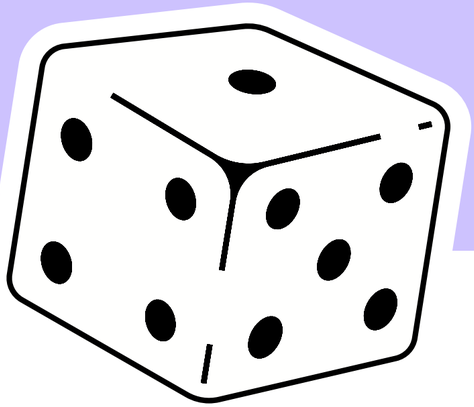
Scripting and Programming

Game logic is implemented using programming languages:

- *Unity: C#*
- *Unreal Engine: Blueprints (visual scripting) and C++*
- *Godot: GDScript (Python-like), C#, C++.*







CHALLENGES IN 3D GAME DEVELOPEMENT

CHALLENGES

Performance Optimization:

Balancing visual quality and frame rate.

Cross-Platform Development:

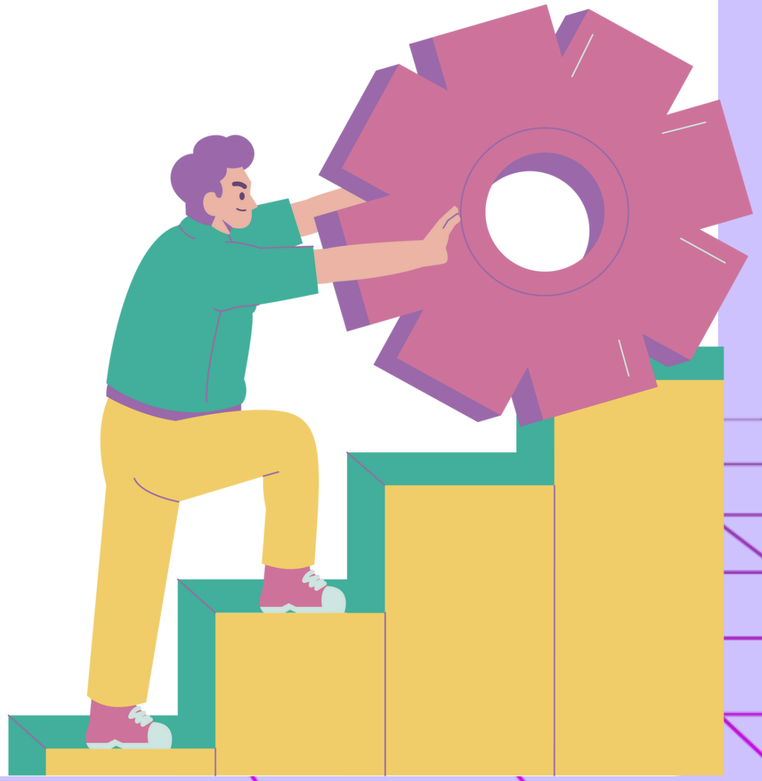
Ensuring compatibility across devices.

Art Pipeline:

Efficiently creating and managing assets.

Learning Curve:

Mastering tools, engines, and programming.



BEST PRACTICES

Start small: Build simple projects before tackling complex games.

Use version control to manage changes.

Test frequently on target platforms.

Learn from existing games and communities (e.g., Unity Forums, Unreal Slackers).







THANK YOU FOR PLAYING!



